
Advanced Computer Architectures

January 20, 2025

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Name
Last Name
ID Number

Problem 1 (20%)	
Problem 2 (15%)	
Problem 3 (15%)	
Question 1 (15%)	
Question 2 (20%)	
Question 3 (15%)	
Total (100%)	

NOTE: students have 90' to complete the exam.

To pass the midterm, a minimum score of 25% is requested by each "section" (section 1: problems, section 2: questions).

The overall score to access the oral has to be greater or equal to 50%

Problem 1

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the **MESI protocol** is used, with **write-back** caches, **write-allocate**, and **write-invalidate** of other caches.

Time	After Operation	Px cache block state	Py cache block state	Pz cache block state	Memory at Block 0 up to date?	Memory at Block 1 up to date?	Memory at Block 2 up to date?
1	Px read block 0	Exclusive (0)	Invalid	Modified (2)	yes	yes	no
2	Py write block 2						
3	Px read block 2						
4	Pz read block 0						
5	Pz write block 2						
6	Px read block 2						
7	Py read block 2						
8	Pz write block 2						
9	Px write block 1						
10	Py read block 1						

Problem 2

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 4, R1 is set to 1)

```
LOOP:      LD      F1  0    R0
           ADDD    F2  F1   F1
           DIVD    F3  F2   F1
           ADDI    R1  R1   1000
LOOP2:     MULTD   F12 F3    F2
           SUBI    R1  R1    2
           BNEZ    R1  LOOP2
           SUBI    R0  R0    4
           BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 3

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	LD \$F1, 0(\$R1)	1	2	6	7
I2	FADD \$F2, \$F2, \$F3	2	3	8	9
I3	ADDI \$R3, \$R3, 8	3	4	5	6
I4	LD \$F4, 0(R2)	4	5	9	10
I5	FADD \$F5, \$F4, \$F2	5	11	16	17
I6	FMULT \$F6, \$F1, \$F4	6	11	21	22
I7	ADDI \$R5, \$R5, 1	7	8	9	10
I8	LD \$R6, 0(\$R4)	8	9	13	14
I9	SD \$F6, 0(\$R5)	11	23	27	28
I10	SD \$F5, 0(\$R6)	15	18	22	23

A. List all the possible conflicts in the code.

B. Is there a SCOREBOARD "configuration" that can respect the shown execution? Yes or No? Please explain why.

C. If the previous table was not correct, please, write the right architectural configuration and specify the number, kind and latency for each unit.

Answer 1.A

Answer 1.B

Answer 1.C

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	LD \$F1, 0(\$R1)				
I2	FADD \$F2, \$F2, \$F3				
I3	ADDI \$R3, \$R3, 8				
I4	LD \$F4, 0(R2)				
I5	FADD \$F5, \$F4, \$F2				
I6	FMULT \$F6, \$F1, \$F4				
I7	ADDI \$R5, \$R5, 1				
I8	LD \$R6, 0(\$R4)				
I9	SD \$F6, 0(\$R5)				
I10	SD \$F5, 0(\$R6)				

Question 1

What is a branch predictor and why/how can it improve the performance of a system?

Answer

Question 2

Please say if the following statements are TRUE or FALSE

Answer 1: the performance of an application is only limited by its code and the intrinsic parallelism

Answer 2: the Scoreboard and Tomasulo architectures can achieve different performance

Answer 3: in coarse-grained multithreading, the processor executes the instructions of a thread until a specific event happens

Answer 4: explicit register renaming requires additional hardware

For each of the answers, motivate why it is TRUE or FALSE

Answer

Question 3

What are structural, data, and control hazards? When do they occur?

Answer

